Fair Test · Knowledge Organiser

Learning goal: I can plan a fair test by naming what to change, what to measure, and what to keep the same.

Today I Can

I can turn a science question into a testable plan.

I can name the one independent variable to change and the dependent variable to measure.

I can list at least three control variables to keep the same and explain why.

Must-Know Words

independent variable: the one factor we change on purpose in a fair test

dependent variable: the factor we measure to see the effect of our change

control variable: a factor we keep the same so the test is fair

fair test: an investigation where only one factor is changed while all others are kept constant

Core Science Steps

First, name the independent variable (what we change): the area of the parachute canopy.

Second, name the dependent variable (what we measure): the fall time in seconds (s).

Third, lock the control variables: same drop height (1.5 m), same 10 g mass, and same strings.

Keep the Science Clear

Use a 3-box variable grid (Change / Measure / Keep the same) before touching any equipment.

Planning variables makes the test fair, but you still need careful release and timing during the practical.